

KIELCE-SUKOW, Poland

WMO: 125700

Lat: 50.8103N

Lon: 20.6922E

Elev: 857

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	8.7	-1.2	5.3	4.1	3.9	6.9	9.8	22.9	39.2	20.5	37.0	3.5	70	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	85.8	68.5	82.4	67.0	79.3	65.3	70.3	82.4	68.6	79.4	67.0	76.3	7.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	99.8	74.6	64.8	94.7	73.0	63.4	90.1	71.5	34.7	82.3	33.3	79.8	31.9	76.3	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.3	14.2	DB	-3.4	90.8	8.4	3.1	-9.5	93.0	-14.5	94.8	-19.2	96.5	-25.3	98.8
			WB	-3.9	73.6	8.3	1.7	-9.9	74.9	-14.8	75.9	-19.4	76.8	-25.5	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	27.3	30.0	36.4	47.1	55.7	61.8	65.5	64.5	56.0	46.8	38.1	30.2
	DBStd	15.42	9.55	9.05	7.91	7.54	6.82	6.28	5.81	5.87	6.41	7.43	7.71	8.73
	HDD50	2999	703	561	423	141	23	1	0	0	18	153	361	615
	HDD65	6879	1168	981	886	538	295	136	67	81	277	564	806	1080
	CDD50	1794	0	0	2	52	200	354	480	450	197	54	5	0
	CDD65	200	0	0	0	0	8	38	82	66	6	0	0	0
	CDH74	2267	0	0	0	22	143	432	875	700	94	1	0	0
	CDH80	578	0	0	0	2	15	99	253	196	13	0	0	0
	WSAvg	6.2	6.9	7.1	7.4	6.8	6.0	5.7	5.5	4.9	5.2	5.8	6.2	6.7
	PrecAvg	24.8	1.5	1.3	1.6	1.6	2.7	2.8	3.9	2.6	2.2	1.6	1.7	1.4
Precipitation	PrecMax	35.1	3.7	3.7	3.1	5.6	5.4	6.9	10.8	5.0	5.4	4.1	3.3	3.3
	PrecMin	19.1	0.3	0.3	0.4	0.0	0.9	0.9	0.1	0.5	0.2	0.2	0.0	0.2
	PrecStd	3.6	0.7	0.7	0.7	1.1	1.3	1.4	2.6	1.1	1.3	1.1	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	53.5	63.5	76.9	82.2	87.3	90.6	90.3	81.6	71.4	59.4	50.2
		MCWB	44.7	46.5	50.9	58.7	64.0	69.7	70.9	69.6	64.4	59.9	53.4	47.0
	2%	DB	44.6	48.6	57.8	70.9	78.0	82.8	86.3	85.0	76.3	67.1	55.4	46.6
		MCWB	41.8	44.0	47.6	55.9	62.6	67.6	68.2	68.6	63.0	58.2	51.0	44.0
	5%	DB	41.8	45.2	53.3	65.9	74.5	78.9	82.5	81.3	72.6	63.1	52.1	43.5
		MCWB	39.6	41.8	45.7	53.2	61.2	65.2	67.3	66.9	61.2	56.3	48.9	41.7
	10%	DB	39.0	41.8	48.9	61.7	70.3	75.4	79.0	77.5	68.7	58.8	49.4	40.8
		MCWB	37.3	39.0	43.5	50.9	59.2	63.6	65.5	64.9	59.4	54.2	47.0	39.3
	0.4%	WB	45.7	47.5	51.9	59.8	67.2	71.6	72.6	71.9	66.6	61.8	54.2	47.6
		MCDB	48.0	51.3	60.0	73.9	78.0	83.7	86.2	84.9	77.7	68.8	57.8	49.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.3	44.7	48.9	57.3	64.4	69.0	70.4	70.0	64.4	59.1	51.7	44.2
		MCDB	44.3	48.0	56.0	68.8	74.5	80.0	82.6	82.1	73.3	65.4	54.8	46.1
	5%	WB	39.7	41.9	46.4	54.5	62.1	66.8	68.8	68.1	62.3	56.9	49.3	42.0
		MCDB	41.6	44.9	52.0	63.7	72.0	76.4	79.4	78.8	70.4	62.1	51.7	43.5
	10%	WB	37.3	39.2	43.8	52.2	60.1	64.7	67.1	66.2	60.5	54.5	47.0	39.4
		MCDB	38.8	41.6	48.3	60.0	68.9	73.3	75.8	75.1	67.3	58.7	49.2	40.7
Mean Daily Temperature Range	5% DB	MDBR	9.8	12.1	15.4	19.6	20.3	19.8	20.1	20.8	18.4	16.0	10.9	9.4
		MCDBR	10.7	16.7	24.5	28.1	27.1	26.0	26.7	26.8	25.6	23.1	16.1	11.5
		MCWBR	9.0	12.1	15.3	14.6	14.0	12.3	11.2	11.9	13.1	14.5	11.9	9.5
	5% WB	MCDBR	10.4	15.3	22.3	25.2	24.2	23.4	23.2	24.3	22.6	21.6	14.3	10.9
		MCWBR	9.0	11.5	14.7	13.8	13.5	11.8	10.8	11.3	12.5	13.9	11.1	9.3
Clear-Sky Solar Irradiance	taub		0.317	0.342	0.384	0.434	0.423	0.421	0.445	0.437	0.403	0.371	0.349	0.313
	taud		2.249	2.239	2.197	2.123	2.210	2.247	2.194	2.218	2.273	2.330	2.320	2.306
	Ebn at Noon		218	245	255	255	264	266	256	252	246	232	205	201
	Edh at Noon		23	30	37	45	43	42	43	40	34	27	21	18
All-Sky Solar Radiation	RadAvg		251	489	861	1295	1612	1728	1623	1458	1037	610	281	190
	RadStd		46	86	111	158	180	144	195	108	152	86	30	29

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.86	N/A	N/A	+1.64	N/A	N/A	N/A	N/A	+103	+50	
(47)	Regional (32 neighbors)	+0.92	N/A	N/A	+1.49	+0.74	+0.67	N/A	-290	+126	+49	

Nomenclature: See separate page